

Problem Statement

Current State:

Facility managers face challenges in efficiently creating and managing staff rosters that comply with labor laws, account for regional holidays, and adapt to changing operational needs. Manual processes are error-prone, time-consuming, and often result in non-compliance or suboptimal shift coverage.

Impact & Affected Stakeholders:

- Facility managers
- Warehouse supervisors
- Ground staff

Why Now:

Increasing regulatory scrutiny and operational complexity make automated, compliant roster management essential for modern facilities.

Quantitative Scope:

Manual roster creation can lead to up to 20% scheduling errors and 15% overtime costs due to non-optimized shifts and non-compliance penalties.

Proposed Solution

Approach:

Develop a web-based application that leverages AI to generate compliant and optimized staff rosters, with CRUD operations for staff profiles and shift assignments. The system will integrate labor law rules and regional holiday calendars to ensure compliance.

Key Decisions:

- Use AI algorithms to automate roster generation based on guidelines and constraints.
- Implement CRUD (Create, Read, Update, Delete) operations for staff and shifts.
- Incorporate labor law and holiday data into scheduling logic.
- Design for extensibility to support additional facility types in the future.

Trade-offs:

- Initial POC will focus on basic features, deferring advanced analytics and mobile support.
- AI recommendations may require manual overrides for edge cases, prioritizing compliance over full automation.
- Regional customization may be limited in the POC phase.

Target Users & Personas

Priya Sharma _(Facility Manager)_

Goals:

- Ensure all shifts are covered efficiently

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- Maintain compliance with labor laws
- Reduce time spent on roster creation

Pain Points:

- Manual scheduling errors
- Difficulty tracking staff availability
- Risk of non-compliance penalties

Behavior Patterns:

- Checks rosters daily
- Updates shifts based on staff requests
- Relies on spreadsheets or paper for current process

Ravi Patel _(Ground Staff)_

Goals:

- Know assigned shifts in advance
- Request shift changes easily
- Avoid overtime violations

Pain Points:

- Unclear shift assignments
- Last-minute changes
- Unpaid overtime

Behavior Patterns:

- Checks shift schedule weekly
- Requests changes via supervisor
- Prefers mobile notifications

User Stories & Use Cases

US1 — Priority: High

As a **Facility Manager**, add a new staff member with basic profile details, so I can include them in future rosters

US2 — Priority: High

As a **Facility Manager**, update a staff member's shift assignment, to accommodate last-minute changes

US3 — Priority: Medium

As a **Facility Manager**, delete a staff member who has left the organization, to keep the roster accurate

US4 — Priority: High

As a **Facility Manager**, generate a weekly roster that complies with labor laws and holidays, to avoid legal penalties and ensure fairness

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US5 — Priority: High

As a **Facility Manager**, receive AI suggestions for optimal shift allocation, to minimize overtime and maximize coverage

US6 — Priority: High

As a **Ground Staff**, view my upcoming shifts, so I can plan my week

US7 — Priority: High

As a **Facility Manager**, be alerted if a roster violates labor law constraints, to correct issues before publishing

US8 — Priority: Medium

As a **Facility Manager**, assign shifts during a regional holiday, to ensure only available staff are scheduled

US9 — Priority: High

As a **Facility Manager**, attempt to assign a staff member to overlapping shifts, to receive an error and prevent double-booking

US10 — Priority: Medium

As a **Ground Staff**, request a shift change, to accommodate personal needs

Functional Requirements

Must Have:

- **FR1:** Ability to add, update, and delete staff member profiles with fields: name, contact, role, and employment status.
- **FR2:** Ability to create, update, and delete shift assignments for staff members.
- **FR3:** AI-assisted roster generation that considers labor law constraints (e.g., maximum weekly hours, mandatory breaks) and regional holidays.
- **FR4:** Validation to prevent overlapping shifts for a single staff member.
- **FR5:** Roster compliance checker that flags violations of labor laws or holiday rules before publishing.
- **FR6:** Staff-facing interface to view assigned shifts.

Should Have:

- **FR7:** Ability for staff to request shift changes or swaps, with manager approval workflow.
- **FR8:** Audit log of roster changes for compliance tracking.

Nice to Have:

- **FR9:** Export roster to CSV or PDF for offline use.
- **FR10:** Push/email notifications to staff for new or changed shifts.

Non-Functional Requirements

Performance:

Roster generation and compliance checks must complete within 5 seconds for up to 200 staff members.

Security:

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All data must be encrypted in transit and at rest. Only authorized users can access or modify roster data.

Scalability:

System should support up to 500 staff members and 10 concurrent managers in the POC phase.

Accessibility:

Web interface must meet WCAG 2.1 AA standards for accessibility.

Reliability:

System uptime must be at least 99% during business hours (8am-8pm local time).

Scope & Constraints

In Scope:

- Staff profile management
- Shift assignment and roster generation
- AI-assisted compliance with labor laws and holidays
- Basic staff and manager interfaces

Out of Scope:

- Payroll integration
- Mobile app development
- Advanced analytics or reporting
- Integration with external HR systems

Constraints:

- Regional holiday and labor law data must be provided or uploaded by the facility manager.
- POC will support English language only.
- User authentication will be basic (username/password) for POC.

Success Metrics & KPIs

Roster creation time reduction

- Baseline: 60 minutes per week (manual process)
- Target: 15 minutes per week
- Measurement: Time tracking of roster creation tasks
- Timeline: Within 3 months of POC launch

Compliance violation rate

- Baseline: 10% of rosters have violations
- Target: 0% violations in published rosters
- Measurement: Automated compliance checker logs
- Timeline: Ongoing

Staff satisfaction with shift visibility

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- Baseline: N/A (new feature)
- Target: 80% positive feedback in staff survey
- Measurement: Quarterly staff survey
- Timeline: Within 6 months of POC launch

Roster accuracy (no overlapping shifts)

- Baseline: 5% overlap errors
- Target: 0% overlap errors
- Measurement: System validation logs
- Timeline: Ongoing

Dependencies & Risks

Dependencies:

- Accurate labor law and regional holiday data
- User authentication system
- AI algorithm for roster optimization

Risks:

- **Incomplete or outdated labor law/holiday data leads to non-compliant rosters**

- Likelihood: Medium | Impact: High

- Mitigation: Require managers to review and update data regularly; provide data upload interface

- **AI-generated rosters may not account for all operational nuances**

- Likelihood: Medium | Impact: Medium

- Mitigation: Allow manual overrides and provide clear compliance warnings

- **User resistance to adopting new system**

- Likelihood: Medium | Impact: Medium

- Mitigation: Provide onboarding materials and support

- **Security breach of staff data**

- Likelihood: Low | Impact: High

- Mitigation: Implement encryption and access controls

Acceptance Criteria

AC1: Add staff member

- GIVEN a facility manager is on the staff management page
- WHEN they enter valid profile details and submit
- THEN:

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- the new staff member appears in the staff list within 2 seconds
- the staff member is available for shift assignment

AC2: Update shift assignment

- GIVEN a staff member is assigned to a shift
- WHEN the manager updates the shift timing
- THEN:
 - the staff member's shift reflects the new timing
 - no overlapping shifts are created

AC3: Delete staff member

- GIVEN a staff member is in the system
- WHEN the manager deletes the staff member
- THEN:
 - the staff member is removed from all future rosters
 - historical roster data is retained for compliance

AC4: AI-assisted roster generation

- GIVEN the manager initiates roster generation for a week
- WHEN AI processes staff availability, labor laws, and holidays
- THEN:
 - a roster is generated within 5 seconds
 - no labor law or holiday violations are present

AC5: Compliance violation alert

- GIVEN a roster contains a shift that violates labor law
- WHEN the manager attempts to publish the roster
- THEN:
 - the system displays a clear error message indicating the violation
 - the roster cannot be published until corrected

AC6: Staff views assigned shifts

- GIVEN a staff member logs into the system
- WHEN they navigate to their schedule page
- THEN:
 - all upcoming shifts for the next 4 weeks are displayed
 - shifts reflect the latest published roster

AC7: Prevent overlapping shifts

- GIVEN a manager attempts to assign overlapping shifts to a staff member

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- WHEN the assignment is submitted
- THEN:

- the system rejects the assignment
- an error message explains the overlap

AC8: Shift assignment on regional holiday

- GIVEN a manager assigns a shift on a regional holiday
- WHEN the assignment is submitted
- THEN:

- the system checks staff availability
- only available staff are assigned
- a warning is shown if coverage is insufficient

AC9: Staff requests shift change

- GIVEN a staff member is assigned to a shift
- WHEN they request a change via the interface
- THEN:

- the manager receives a notification
- the request is logged for approval

AC10: Export roster

- GIVEN a manager views the finalized roster
- WHEN they select 'Export to CSV'
- THEN:

- the roster is downloaded as a CSV file within 3 seconds
- all shift and staff data is included

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